CHAPTER 2 / SUPPLEMENTARY MATERIAL

Cyber–Physical Security and Mission Assurance for Unmanned Aerial Vehicles

Airspace Cybersecurity: Attack Taxonomy and Real-World Incidents

Companion figures: threat classification, passive observation, navigation interference, and multi-UAV coordination

Conceptual scenarios support the chapter discussion. These figures do not document specific incidents.

Airspace attack taxonomy

Threat classes overlap. A single event can affect several interfaces and mission functions.

Threat class	Exposed interface / examples	Potential mission consequence
Passive observation	RF links, metadata, imagery Eavesdropping and traffic analysis	Disclosure of data or operating patterns
Communication disruption	C2, telemetry, inter-UAV links Interference and denial of service	Delayed commands, lost links, or reduced coordination
Navigation deception	GNSS receiver and PNT inputs Spoofing and meaconing / replay	Incorrect position or time, route deviation
System or data compromise	Mission plans, software, shared state Waypoint or telemetry manipulation	Unauthorized behavior or corrupted situational awareness
Swarm coordination attacks	Leader, membership, consensus Impersonation and false shared data	Formation errors, inconsistent decisions, or mission interruption

Classification guide: the interface, mechanism, and observed consequence should be recorded separately.

C2: command and control PNT: positioning, navigation, and timing

Passive RF and sensor observation

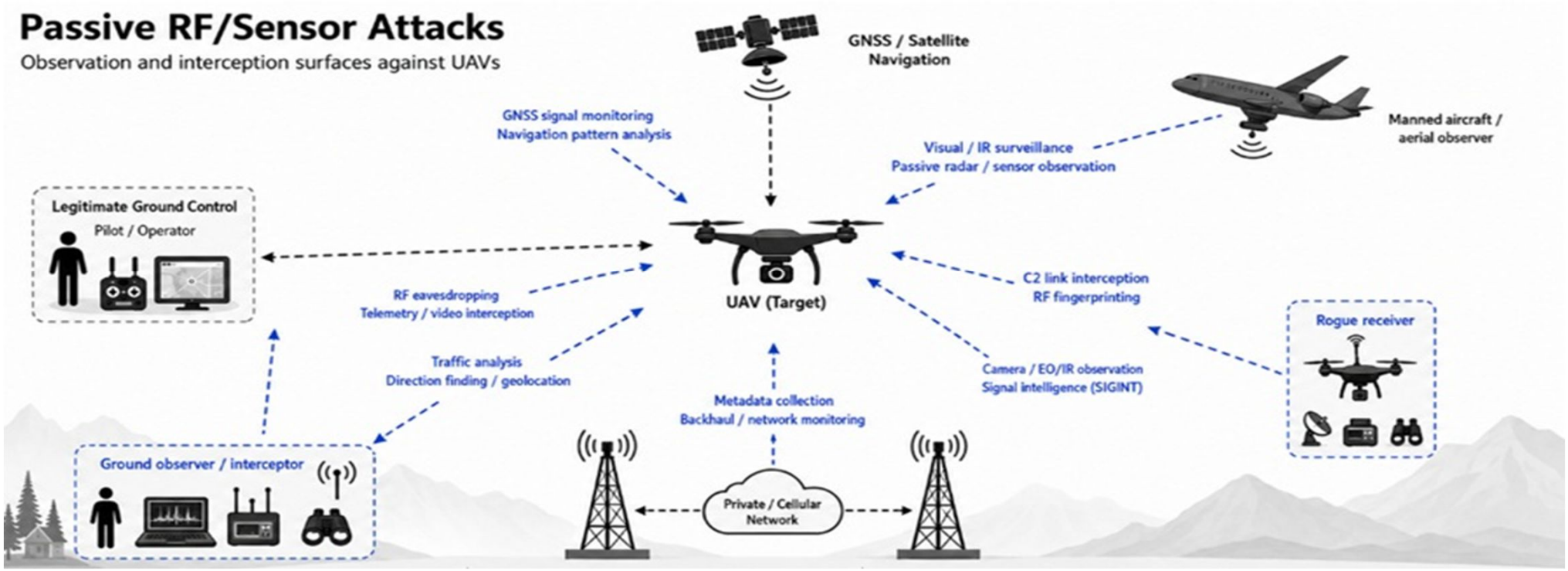


Figure S2.1. Passive collection can reveal communications, location, and operating patterns. Observation alone does not demonstrate control of the aircraft.

Black dashed links: legitimate signals. Blue dashed paths: conceptual observation or interception.

GNSS interference and navigation manipulation

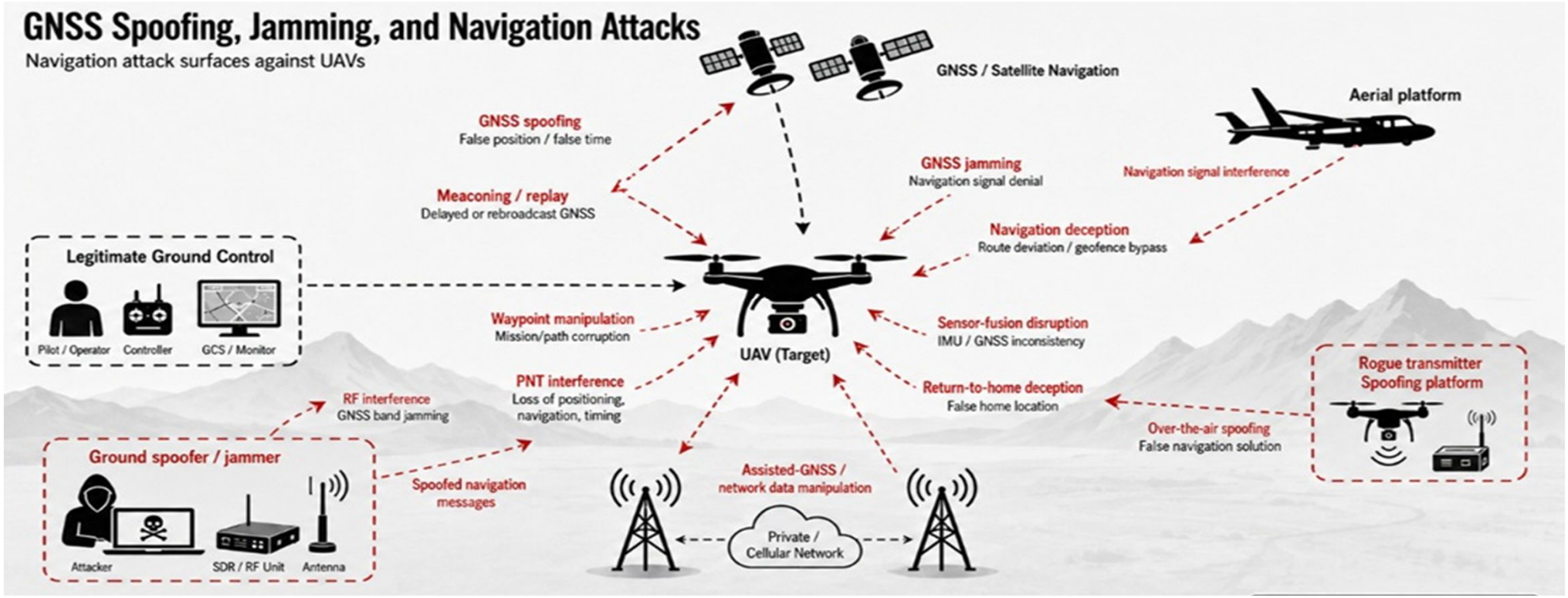


Figure S2.2. Jamming can deny GNSS reception. Spoofing can corrupt position or time. Waypoint manipulation requires a separate compromised interface.

Black dashed links: legitimate signals. Red dashed paths: conceptual interference or manipulation.

Swarm and multi-UAV coordination threats

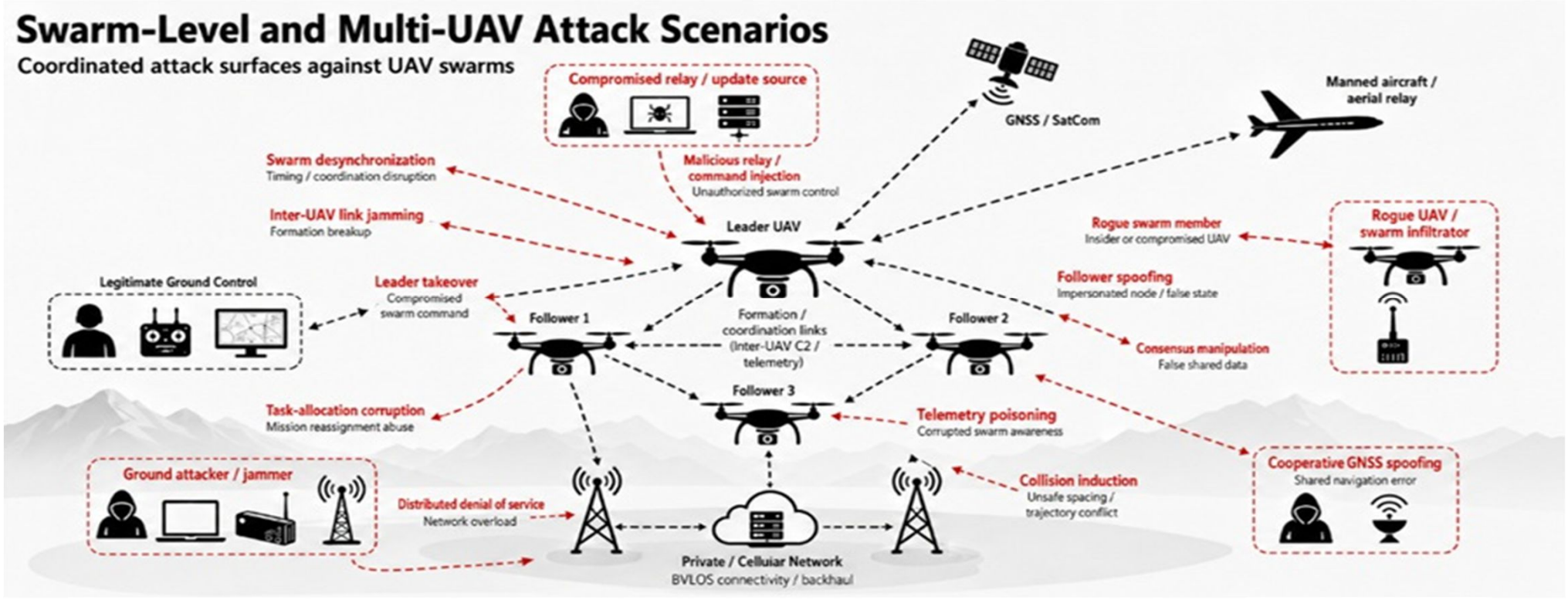


Figure S2.3. Corrupted links, identities, or shared state can affect formation and task allocation. The impact depends on the coordination protocol and available fallback behavior.

Black dashed links: legitimate coordination. Red dashed paths: conceptual threat relationships.